Lab - 2

PAGE :
DATE : 1/10/26

Genetic Algorithm

Pseudocode:

Start

1) Define the fitness function
2) Declare - size_of_population
   mutation_rate
   crossover_rate
3) Initialize: a population randomly
4) FOR the population:
   i) Apply fitness function.
   ii) Select best individuals based on fitness function.
   iii) Crossover of best individuals to produce offsprings.
   iv) Replace the old population with new offsprings.
   v) Introduce mutations in the offsprings to induce changes.
   vi) Select the fittest individual.
   vii) Display the fittest individual.
5) END

TSP

IEEE paper

PAGE :
DATE : / /

Result :- Perturbed GA beat the plain GA on fitness, convergence rate, error rate and average convergence. d = 5%. gave the best results.

Formula.

$c'(e) = v \times (c(e))$

$c(e)$ → True distance of one edge e taken from distance matrix.

v is random multiplier drawn seperately for each edge.

$c'(e)$ is perturbed (noisy) distance the GA uses to judge tours.

d is the noise size. (largest fractional change an edge can get)

Process

coding : complicated

Ctrl

1/10/26

IEEE Title

PAGE :
DATE : / /

Travelling Salesman Problem solving using Genetic Algorithm combined Data Perturbation.

Summary:

The Travelling Salesman Problem (TSP) asks for the shortest tour that visits every city exactly once and returns to the start. It is NP-hard, so exact methods become impractical as the number of cities grows.

Weakness of a plain GA: A genetic algorithm handles TSP style problems well but has weak local search. It tends to converge quickly to a local optimum and loses population diversity.

Proposed fix (perturbed GA): Before and during evolution, the distance matrix is slightly distorted with random noise:
$C'(e) = V \times C(e)$, where $V \sim$ Uniform $(1-d, 1+d)$
d - controls how large the noise is.

Experiment:-
The authors tested on six TSPLIB instances (eil51, st70, KroA100, KroC100, lin105, tsp225) comparing GA against perturbed GA with $d = 5\%, 10\%, 20\%$. each setup ran 25 times using matlab.